Insights Report

Advertising Funnel Analysis

Impression	Click	Engagement	Purchase
9,594	9,423	8,050	1,778

1️⃣ Executive Summary

This report presents analytical findings from a comprehensive funnel analysis of a digital marketing agency managing paid social media campaigns across Facebook and Instagram. The analysis covers the full user journey — impression to purchase — answering 14 analytical questions through SQL queries and Power BI dashboards built on a custom relational database of 300,000+ ad events.

Key Strategic Findings

- Facebook outperforms Instagram in conversion rate (~10% vs ~7%)
- Stories ads drive the highest conversion at 6.6% — 94% above video ads
- Engagement → Purchase is the critical bottleneck with a 77.91% drop-off
- Technology, Gaming, and Art audiences convert at the highest rates (19–20%)
- Thursday mornings are the single best time slot to run ads
- Average time to purchase is 43 days — campaigns need long retargeting windows

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2️⃣ Funnel Overview & Conversion Analysis

Funnel Stage Performance

The top of the funnel is highly efficient — 98.22% of impressions turn into clicks — but the bottom leaks badly at the Engagement → Purchase step.

Impression → Click	**98.22%**
Click → Engagement	**85.43%**
Engagement → Purchase	**22.09%**
Overall (Impression → Purchase)	**18.53%**

Critical Drop-Off: Engagement → Purchase

The biggest drop-off is between Engagement and Purchase, where 77.91% of engaged users fail to convert. This suggests friction in the post-click journey, insufficient purchase intent at the engagement stage, or a lack of retargeting to re-engage warm users.

Recommendations to address the drop-off

- Launch retargeting campaigns specifically for users who engaged but did not purchase
- A/B test landing pages to reduce post-click friction
- Introduce urgency triggers (limited-time offers) to nudge engaged users toward purchase
- Review alignment between ad content and landing page messaging

Top-of-Funnel Strength

Measured from impressions, 98.22% of users reach click, 83.91% reach engagement, and 18.53% reach purchase. The high impression-to-click rate confirms that targeting is effective — the challenge is purely at the bottom of the funnel.

3️⃣ Campaign & Ad Format Performance

Top 5 Campaigns by conversion rate

Campaign	Conversion rate (Imp. → Purchase)
campaign_38_q3	**1.28%**
campaign_17_launch	1.25%
campaign_33_summer	1.19%
campaign_27_q3	1.14%
campaign_24_summer	1.12%

Q3 campaigns appear twice in the top 5, suggesting that Q3 strategies or seasonal timing consistently produce stronger results.

Platform Comparison

Platform	Conversion	Takeaway
Facebook	**~10.5%**	Primary conversion platform
Instagram	~7.0%	Better for awareness/engagement

Facebook delivers approximately 50% higher conversion rates. Budget allocation should be weighted toward Facebook while Instagram remains a valuable top-of-funnel channel.

Ad Type Performance

Ad Type	Conversion
Stories	**6.6%**
Carousel	5.1%
Image	4.8%
Video	**3.4%**

📌**Ad format recommendation**

- Prioritize Stories as the primary format for conversion-focused campaigns
- Rethink video creative — despite higher production cost, it underperforms all formats
- Consider testing video content reformatted as vertical Stories

Engagement vs Purchase

Engaged-purchase rate
18.60%

Non-Engaged -purchase rate
18.20%

Insight: engagement metrics are not purchase predictors
The 0.4pp gap is negligible across ~10,000 users. Social engagement (likes, comments, shares) does not meaningfully drive purchase decisions here. Campaign KPIs should focus on purchase-intent signals, not engagement volume.

4️⃣ Audience Analysis

Gender

Gender	Conversion
Female	18.84%
Male	18.39%
Other	18.28%

Only a 0.56pp spread across all groups — gender alone should not drive targeting decisions, but gender-specific creative messaging may still improve performance.

Age Groups

Age group	Conversion rate
35–44	**19.14%**
45–54	**19.09%**
18–24	18.82%
25–34	18.46%
55–65	18.18%
16–17	16.69%

The 35–54 bracket converts highest. The 16–17 group is the weakest — this segment should be targeted with brand awareness objectives, not direct conversion.

Top 5 interests by conversion rate

Interest	Conversion
Technology	**19.9%**
Gaming	19.1%
Art	19.0%
Finance	18.9%
Food	18.8%

🎯 **Audience targeting recommendation**
Combine the 35–54 age bracket with Technology and Gaming interest targeting for the highest-probability conversion audience. Technology and Gaming convert ~1.7 points above the overall average.

5️⃣ Temporal Patterns & Timing

Average time to purchase
43 days

Slowest Conversion
90 days

Fastest conversion
0 days

The 43-day average means short-burst campaigns miss the majority of converters. Retargeting windows should span 45–90 days. Day-0 converters are high-intent users — serve them fast-follow conversion ads. Long-tail users (up to day 90) respond to urgency-based or reminder messaging.

Day of week

Day	Conversion
Thursday	**13.1%**
Friday	12.1%

Day	Conversion
Monday	11.9%
Saturday	11.7%
Sunday	11.0%
Wednesday	10.8%
Tuesday	10.0%

Time of Day

Time	Conversion
Morning	13.5%
Night	12.7%
Afternoon	12.4%
Evening	12.4%

⏰ **Timing recommendation**

- Schedule highest-budget delivery on Thursday mornings
- Reduce or test different creative on Tuesdays
- Set 45–90 day retargeting windows to align with the 43-day purchase journey

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6️⃣ Strategic Recommendations

Priority	Action	Impact
High	Launch retargeting for engaged non-purchasers	Reduce 77.91% drop-off
High	Reallocate budget toward Facebook over Instagram	~50% higher conversion ROI

Priority	Action	Impact
High	Shift video ad spend to Stories format	+3.2 % conversion lift
Medium	Target 35–54 age + Technology/Gaming interests	Higher audience quality
Medium	Schedule delivery on Thursday mornings	~3% conversion uplift
Low	Extend retargeting windows to 45–90 days	Capture late users

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7️⃣ Methodology & Technical Notes

Data Architecture

Star-schema relational database with four entities:

- **dim_users** — demographics: gender, age group, interests, location
- **dim_campaigns** — campaign metadata: budget, duration, naming
- **dim_ads** — ad attributes: platform, type, targeting parameters
- **fact_ad_events** — 10,000+ events: impression, click, engagement, purchase per user per ad

Dataset is synthetic, generated with Python (Faker + NumPy) to simulate realistic ad campaign structure without personal data.

Analytical Approach

All questions answered using MySQL with CTEs for multi-step funnel calculations. Key techniques:

- DISTINCT user counting at each funnel stage to avoid double-counting
- NULLIF guards on all division operations to handle zero-denominator cases
- DATEDIFF-based time-to-purchase joining first impression dates to first purchase dates

- SUBSTRING_INDEX string splitting to normalize multi-value interest fields
- UNION ALL patterns to build step-to-step and top-of-funnel conversion tables in a single query

Visualization

Results visualized in Microsoft Power BI across four thematic dashboards: Funnel Overview, Campaign Performance, Audience Analysis, and Temporal Patterns.

🛠️ **Tools Used**
MySQL · Power BI · Python (Pandas, NumPy) · Star Schema · CTEs · Window Functions